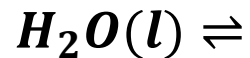
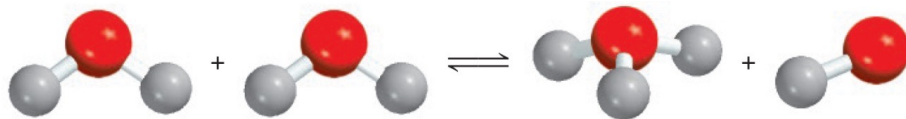
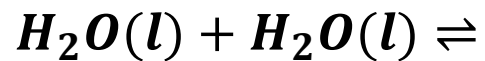


Learning Objectives

1. Calculate the pH, pOH, $[\text{H}_3\text{O}^+]$, and $[\text{OH}^-]$ of a solution when the value of one is known (i.e. calculate the $[\text{H}_3\text{O}^+]$ and $[\text{OH}^-]$ when the pH is known)
2. Rank the relative strengths of weak acids based on K_a or $\text{p}K_a$.
3. Calculate the pH of a strong acid or strong base dissolved in water.
4. Calculate the pH, percentage ionization, and equilibrium concentrations for weak acids or weak bases dissolved in water.
5. Write K_a or K_b constant expressions and write the ionization reaction that corresponds to a specific K_a or K_b value for a weak acid or base.
6. Calculate the K_a value or K_b value when given either:
 - The initial concentration of weak acid or base and pH
 - The initial concentration of weak acid or base and % ionization
 - Equilibrium concentrations

Autoionization of Water

- Water is amphoteric:

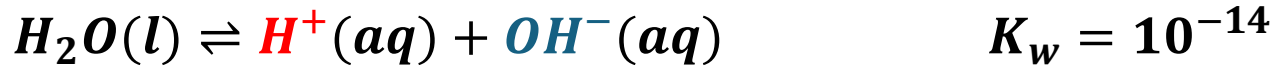


- Ion product constant (____):

$$K_w =$$

Autoionization of Water

- At 25 °C, $K_w =$ _____.
- In pure water, at 25 °C, $[H^+] = [OH^-] =$ _____.



I
C
E

- H^+ and OH^- _____ be present in the same solution at the same time
- $[H^+]$ and $[OH^-]$ _____ on each other, and whichever is the majority will dominate the property of the solution.

- At 25 °C, $[OH^-] =$ _____

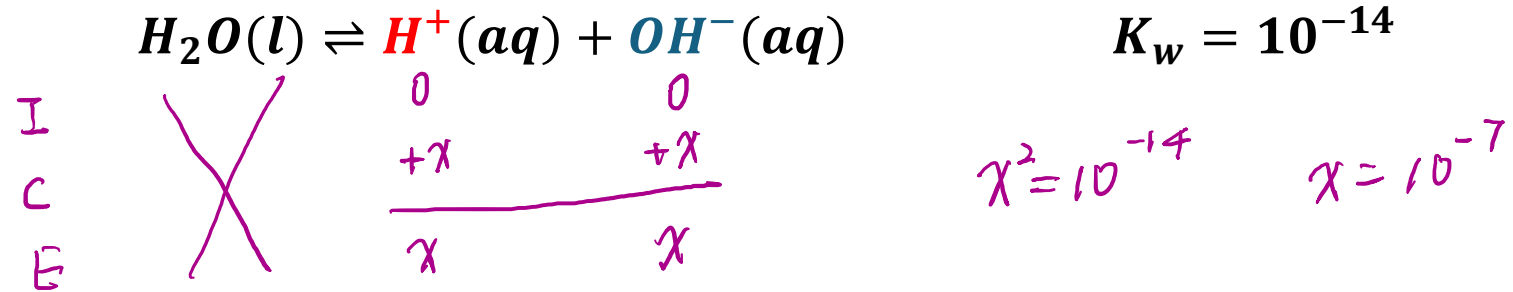
when $[H^+] = 10^{-7} M$, $[OH^-] =$ _____, $[H^+] \text{ } [OH^-]$. the solution is _____

when $[H^+] > 10^{-7} M$, $[OH^-] \text{ } [H^+] \text{ } [OH^-]$. the solution is _____

when $[H^+] < 10^{-7} M$, $[OH^-] \text{ } [H^+] \text{ } [OH^-]$. the solution is _____

Autoionization of Water

- At 25 °C, $K_w = 10^{-14}$.
- In pure water, at 25 °C, $[H^+] = [OH^-] = 10^{-7} M$.



- H^+ and OH^- can be present in the same solution at the same time
- $[H^+]$ and $[OH^-]$ depend on each other, and whichever is the majority will dominate the property of the solution.

$$[H^+][OH^-] = K_w$$

- At 25 °C, $[OH^-] = \frac{K_w}{[H^+]}$

when $[H^+] = 10^{-7} M$, $[OH^-] = 10^{-7} M$, $[H^+] = [OH^-]$. the solution is neutral

when $[H^+] > 10^{-7} M$, $[OH^-] < 10^{-7} M$, $[H^+] > [OH^-]$. the solution is acidic

when $[H^+] < 10^{-7} M$, $[OH^-] > 10^{-7} M$, $[H^+] < [OH^-]$. the solution is basic

Your turn!

The $[OH^-]$ in a certain household cleaning product is 0.0025 M.

1. Assume that the temperature is 25 °C, is this solution acidic, acidic or neutral.
2. Calculate $[H^+]$ in this solution.



$$4.0 \times 10^{-12} M$$

Your turn!

The $[OH^-]$ in a certain household cleaning product is 0.0025 M. $= 2.5 \times 10^{-3} M$

1. Assume that the temperature is 25 °C, is this solution acidic, basic or neutral.

basic

2. Calculate $[H^+]$ in this solution.

$$[H^+] = \frac{K_w}{[OH^-]} = \frac{10^{-14}}{2.5 \times 10^{-3}} M = \underline{4.0 \times 10^{-12} M}$$



$$4.0 \times 10^{-12} M$$

Announcement 5/1

Assignments that are due soon

- Pre-Class Assignment Week 14-M

Office Hour

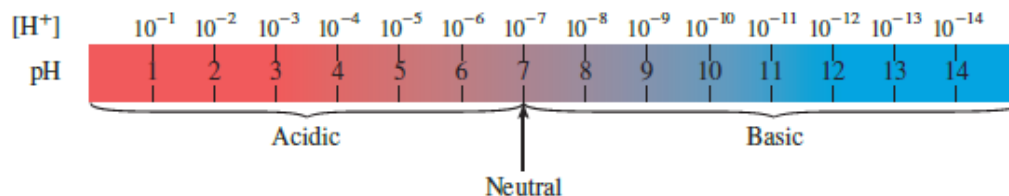
- Dr. Wang's: Fri 2-3 PM at KC 382



pH and pH Scale

- $pH = -\log[H^+]$; $[H^+] = 10^{-pH}$
 - $[H^+]$ is in the unit of M
 - pH has no unit
- If $[H^+] = 0.001\text{ M}$, $pH = -\log(10^{-3}) = +3$
- If $pH = 10$, $[H^+] = 10^{-10}\text{ M}$

- the smaller the pH , the larger the $[H^+]$, and the more acidic the solution is
 - At 25 °C $K_w = 10^{-14}$
 - when $[H^+] = [OH^-]$, the solution is neutral; $[H^+] = 1.0 \times 10^{-7}$ and $pH = 7$
 - when $[H^+] > [OH^-]$, the solution is acidic; $[H^+] > 1.0 \times 10^{-7}$ and $pH < 7$
 - when $[H^+] < [OH^-]$, the solution is basic; $[H^+] < 1.0 \times 10^{-7}$ and $pH > 7$
- at higher $K_w = 10^{-12}$ neutral $pH = 6$



pH and pH Scale

- $pH = -\log[H^+]$; $[H^+] = 10^{-pH}$
- $[H^+]$ is in the unit of M or $\frac{mol}{L}$
- pH has no unit

$$\log_{10}[H^+] = -pH$$

$$[H^+] = 10^{-pH}$$

If $[H^+] = 0.001 M$, $pH = -\log(10^{-3}) = -(-3) = 3$

If $pH = 10$, $[H^+] = 10^{-10} M$

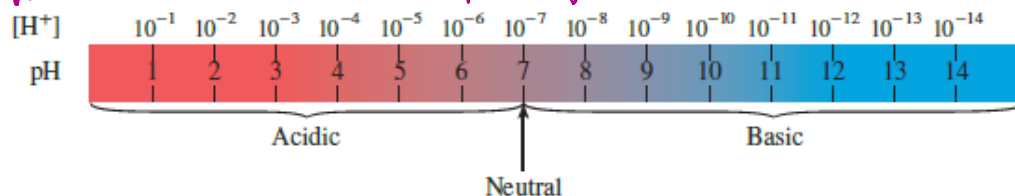
- the smaller the pH , the larger the $[H^+]$, and the more acidic the solution is
- At 25 °C

when $[H^+] = [OH^-]$, the solution is neutral; $[H^+] = 1.0 \times 10^{-7}$ and $pH = 7$

when $[H^+] > [OH^-]$, the solution is acidic; $[H^+] > 1.0 \times 10^{-7}$ and $pH < 7$

when $[H^+] < [OH^-]$, the solution is basic; $[H^+] < 1.0 \times 10^{-7}$ and $pH > 7$

(At 100°C, $K_w = 5.13 \times 10^{-13}$ and pH of neutral solution ($[H^+] = [OH^-] = 6.1$) $= 7.2 \times 10^7$)



pH and pH Scale

The pHs of Some Common Fluids

Sample	pH Value
--------	----------

Gastric juice in the stomach	1.0–2.0
------------------------------	---------

Lemon juice	2.4
-------------	-----

Vinegar	3.0
---------	-----

Grapefruit juice	3.2
------------------	-----

Orange juice	3.5
--------------	-----

Urine	4.8–7.5
-------	---------

Water exposed to air*	5.5
-----------------------	-----

Saliva	6.4–6.9
--------	---------

Milk	6.5
------	-----

Pure water	7.0
------------	-----

Blood	7.35–7.45
-------	-----------

Tears	7.4
-------	-----

Milk of magnesia	10.6
------------------	------

Household ammonia	11.5
-------------------	------

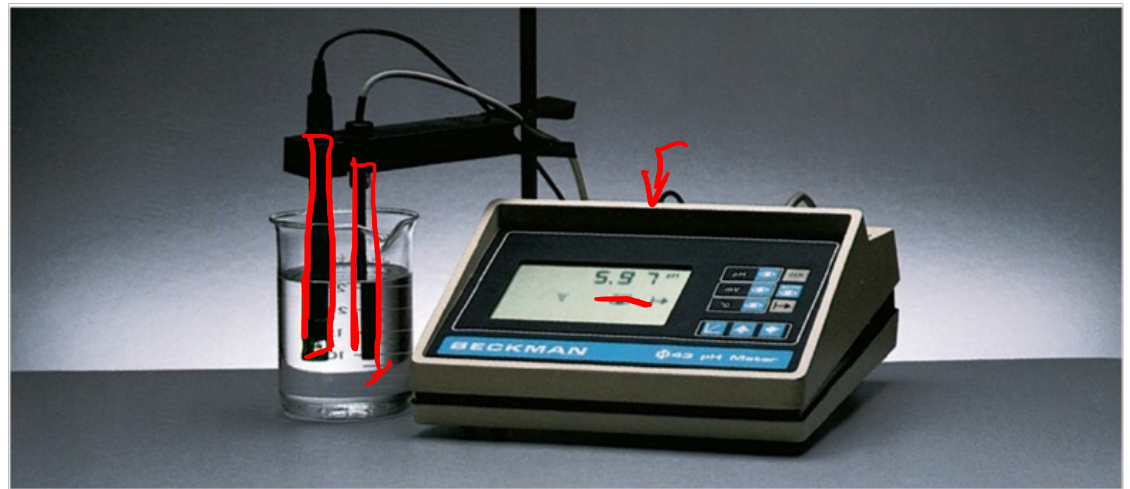
pH test strips



pH indicators

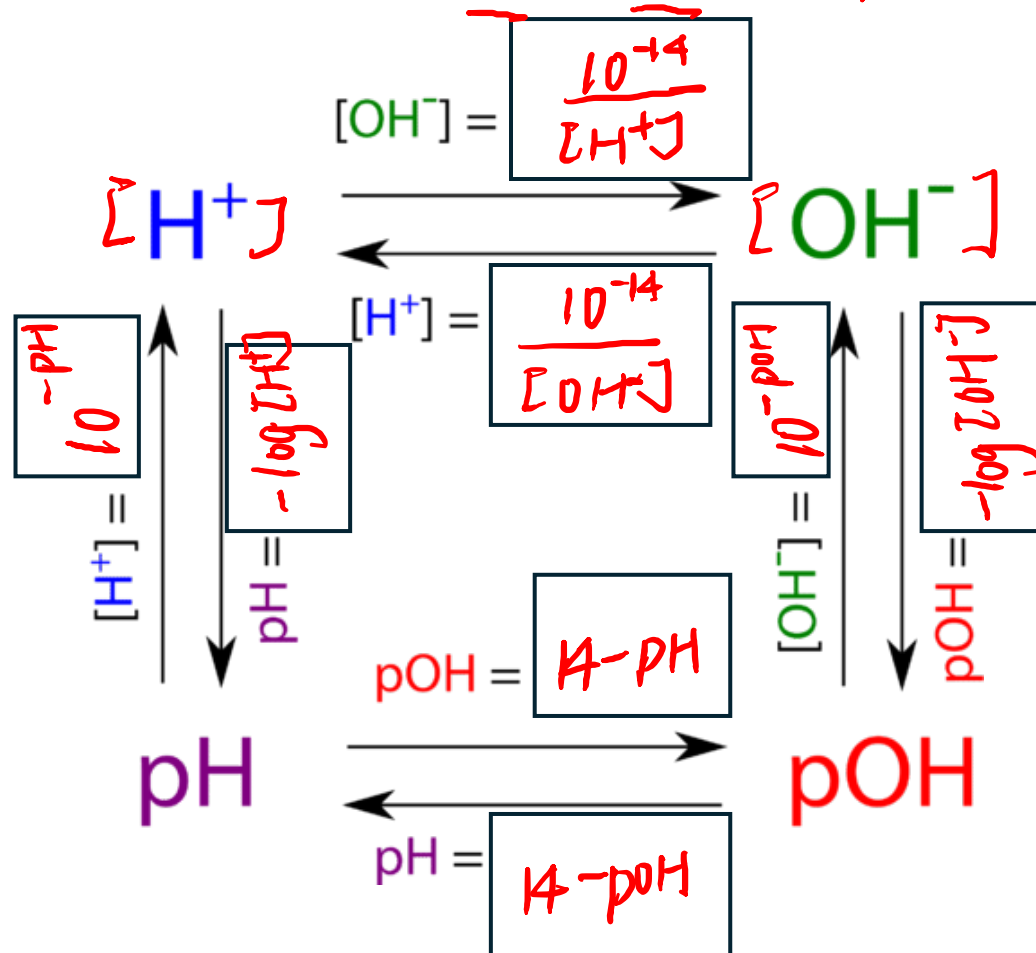


pH meter



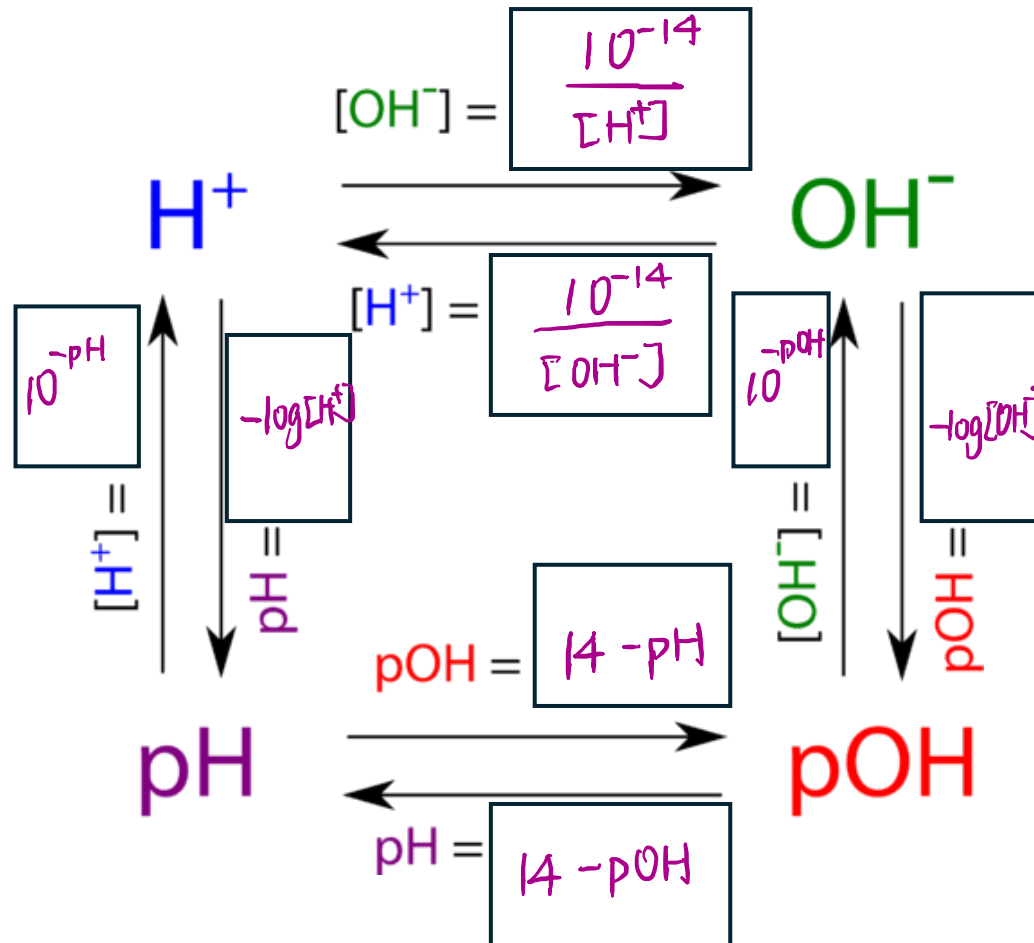
pH, pOH and pH Scale

- $pOH = -\log[OH^-]$; $[OH^-] = 10^{-pOH}$
- At 25 °C, since $K_w = [H^+][OH^-] = 10^{-14}$
- $$-\log([H^+][OH^-]) = (-\log[H^+]) + (-\log[OH^-]) = -\log(10^{-14}) = 14$$
- $$pH + pOH = 14$$



pH, pOH and pH Scale

- $pOH = -\log[OH^-]$; $[OH^-] = 10^{-pOH}$
- At 25 °C, since $K_w = [H^+][OH^-] = 10^{-14}$
- $\log [H^+] + \log [OH^-] = -14$
- $-\log [H^+] - \log [OH^-] = 14$
- $pH + pOH = 14$



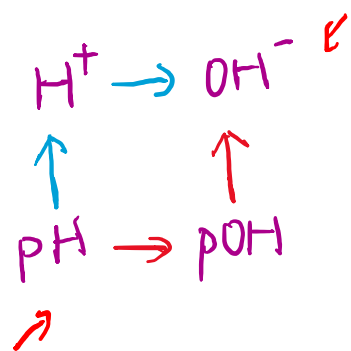
Your turn!

The pH of rainwater collected in a certain region of the northeastern United States on a particular day was 4.82. Assume the temperature is 25 °C, calculate the $[OH^-]$ of the rainwater.

$$6.6 \times 10^{-10} M$$

Your turn!

The $p\text{H}$ of rainwater collected in a certain region of the northeastern United States on a particular day was 4.82. Assume the temperature is 25 °C, calculate the $[\text{OH}^-]$ of the rainwater.



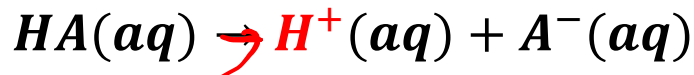
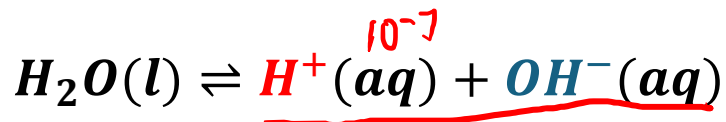
$$\begin{aligned} \textcircled{1} \quad [\text{H}^+] &= 10^{-4.82} = 1.5 \times 10^{-5} \text{ M} \\ [\text{OH}^-] &= \frac{10^{-14}}{1.5 \times 10^{-5}} \text{ M} = 6.6 \times 10^{-10} \text{ M} \end{aligned}$$

$$\textcircled{2} \quad \text{pOH} = 14 - 4.82 = 9.18$$

$$[\text{OH}^-] = 10^{-9.18} = 6.6 \times 10^{-10} \text{ M}$$

pH of Strong Acid/Base Solutions

- Multiple equilibrium exists at the same time.



- The ionization/dissociation of strong acid/base is completion.
- Ignore the contribution from water and determine $[H^+]/[OH^-]$ at equilibrium from strong acid/base.

stoichiometry

$$pH = -\log(10^{-3}) = 3$$

	$HCl(aq)$	$\rightarrow$	$H^+(aq)$	+	$Cl^-(aq)$
Initial (M):	1.0×10^{-3}		0.0		0.0
Change (M):	-1.0×10^{-3}		$+1.0 \times 10^{-3}$		$+1.0 \times 10^{-3}$
Final (M):	0		10^{-3}		10^{-3}

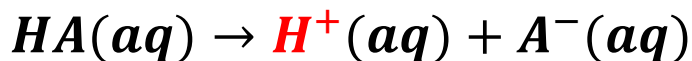
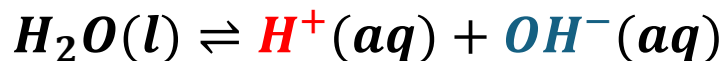
	$Ba(OH)_2(aq)$	$\rightarrow$	$Ba^{2+}(aq)$	+	$2OH^-(aq)$
Initial (M):	0.020		0.00		0.00
Change (M):	-0.020				$+0.020 \times \frac{2}{1}$
Final (M):	0				0.040

$$pOH = -\log(0.040) = 1.4$$

$$pH = 14 - 1.4 = 12.6$$

pH of Strong Acid/Base Solutions

- Multiple equilibrium exists at the same time.



- The ionization/dissociation of strong acid/base is completed.
- Ignore the contribution from water and determine $[H^+]/[OH^-]$ at equilibrium from strong acid/base.

	$HCl(aq)$	$\longrightarrow$	$H^+(aq)$	+	$Cl^-(aq)$
Initial (M):	1.0×10^{-3}		0.0		0.0
Change (M):	-1.0×10^{-3}		$+1.0 \times 10^{-3}$		$+1.0 \times 10^{-3}$
Final (M):	0		1.0×10^{-3}		1.0×10^{-3}

$$[H^+] = 1.0 \times 10^{-3} M$$

$$pH = 3$$

$$[OH^-] = 1.0 \times 10^{-11} M$$

	$Ba(OH)_2(aq)$	$\longrightarrow$	$Ba^{2+}(aq)$	+	$2OH^-(aq)$
Initial (M):	0.020		0.00		0.00
Change (M):	-0.020		$+0.020$		$+0.040$
Final (M):	0		0.020		0.040

$$[OH^-] = 0.040 M$$

$$[H^+] = 2.5 \times 10^{-13} M$$

$$pH = 12.6$$

Your turn!

HBr is a strong acid and what is the pH of 12.0 M of HBr solution?



$$pH = -\log(12.0) = \boxed{-1.08}$$

Your turn!

What is the concentration of $\text{Ca}(\text{OH})_2$ in a solution if the pH is 7.92?

$$4.2 \times 10^{-7} M$$

Your turn!

What is the concentration of $\text{Ca}(\text{OH})_2$ in a solution if the $p\text{H}$ is 7.92?



$$p\text{OH} = 14 - 7.92 = 6.08$$

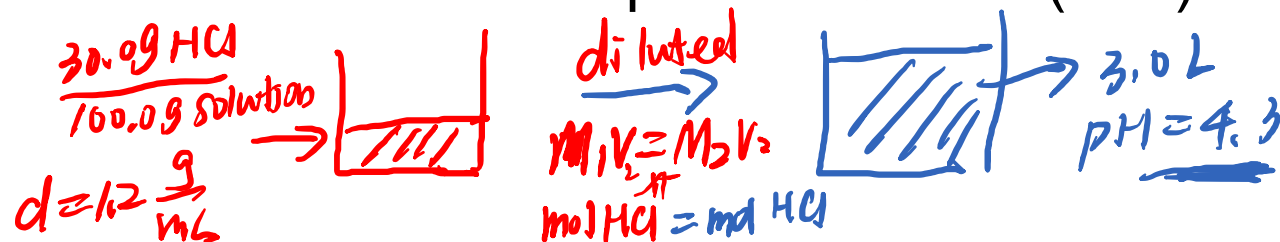
$$[\text{OH}^-] = 10^{-6.08} = 8.32 \times 10^{-7} \text{ M}$$

$$[\text{Ca}(\text{OH})_2] = 8.32 \times 10^{-7} \text{ M } [\text{OH}^-] \times \frac{1[\text{Ca}(\text{OH})_2]}{2[\text{OH}^-]} = 4.2 \times 10^{-7} \text{ M}$$

$$4.2 \times 10^{-7} \text{ M}$$

Your turn!

What volume, in mL, of a 30.0% by mass HCl solution, which has a density of 1.2 g/mL, should be used to make 3.0 L of an HCl solution with a pH of 4.3? $M(HCl) = 36.46 \text{ g/mol}$



pH $\rightarrow$ $[H^+]$ $\rightarrow$ M of HCl $\rightarrow$ mol of HCl

mol HCl $\xleftarrow{M(HCl)}$ mass of HCl $\xleftarrow{\text{mass}\%}$ mass of solution $\xleftarrow{d}$ V of solution

Your turn!

$$\frac{30 \text{ g HCl}}{100 \text{ g solution}}$$

What volume, in mL, of a 30.0% by mass HCl solution, which has a density of 1.2 g/mL, should be used to make 3.0 L of an HCl solution with a pH of 4.3? $M(\text{HCl}) = 36.46 \text{ g/mol}$

$$[\text{H}^+] = 10^{-4.3} \text{ M} = 5.0 \times 10^{-5} \text{ M}$$

$$\frac{5.0 \times 10^{-5} \text{ mol H}^+}{1 \text{ L}} \times 3.00 \text{ L} \times \frac{1 \text{ mol HCl}}{1 \text{ mol H}^+} \times \frac{36.46 \text{ g HCl}}{1 \text{ mol HCl}} \times \frac{100 \text{ g soln}}{30 \text{ g HCl}} \times \frac{1 \text{ mL soln}}{1.2 \text{ g soln}}$$

$$= 0.015 \text{ mL solution}$$

Announcements 5/4

Assignments that are due soon

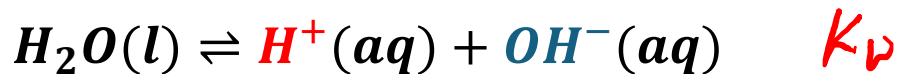
- Pre-Class Assignment Week 14-W
- Case study Assignment

Office Hour

- Dr. Wang's: Tue 10-11 AM at KC 230
- Paige Kwan's: Mon 11:30 AM -12:30 PM at the TLC (Demille 160P)
- Saige Douglass's: Tue 12:00-1:00 PM at the TLC (Demille 160P)

pH of Weak Acid/Base Solutions

- Multiple equilibrium exists at the same time.



- Acid ionization constant (K_a)

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

- The larger the K_a , the stronger the acid.

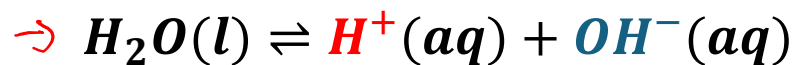
Acidity
stronger

$B(aq) + H_2O(l) \rightleftharpoons HB^+(aq) + OH^-(aq)$
 $K_b = \frac{[HB^+][OH^-]}{[B]}$

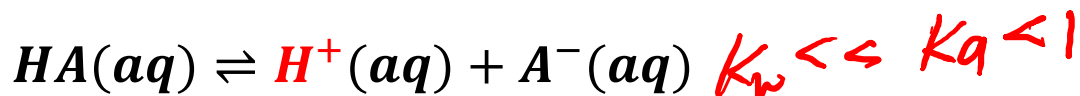
Name of Acid	Formula	Structure	K_a
Hydrofluoric acid	HF	<chem>F</chem>	7.1×10^{-4}
Nitrous acid	<chem>HNO2</chem>	<chem>O=N-O-H</chem>	4.5×10^{-4}
Acetylsalicylic acid (aspirin)	<chem>C9H8O4</chem>	<chem>CC(=O)Oc1ccccc1C(=O)O</chem>	3.0×10^{-4}
Formic acid	<chem>HCOOH</chem>	<chem>O=C(O)</chem>	1.7×10^{-4}
Ascorbic acid*	<chem>C6H8O6</chem>	<chem>O=C1OC(O)C(O)C(O)C1O</chem>	8.0×10^{-5}
Benzoic acid	<chem>C6H5COOH</chem>	<chem>O=C(O)c1ccccc1</chem>	6.5×10^{-5}
Acetic acid	<chem>CH3COOH</chem>	<chem>CC(=O)O</chem>	1.8×10^{-5}
Hydrocyanic acid	<chem>HCN</chem>	<chem>[H]C#N</chem>	4.9×10^{-10}

pH of Weak Acid/Base Solutions

- Multiple equilibrium exists at the same time.

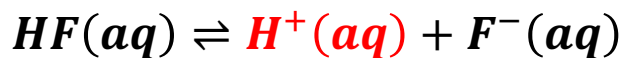


$$K_w = 10^{-14}$$



- Ignore the contribution from water and determine $[\text{H}^+]/[\text{OH}^-]$ at equilibrium from weak acid/base.

Example: Calculate the pH of a 0.50 M HF solution at 25 °C.



$$K_a = \frac{[\text{H}^+][\text{F}^-]}{[\text{HF}]} = 7.1 \times 10^{-4}$$

Initial (M):	0.50	0	0
Change (M):	-x	+x	+x
Equilibrium (M):	0.50-x	x	x

$$\frac{x \cdot x}{0.50 - x} = 7.1 \times 10^{-4}$$

$$\frac{x^2}{0.50 - x} = 7.1 \times 10^{-4}$$

$$\frac{1}{a} x^2 + \frac{7.1 \times 10^{-4}}{b} x - \frac{3.55 \times 10^{-4}}{c} = 0$$

$$x^2 = 7.1 \times 10^{-4} \times 0.50 - 7.1 \times 10^{-4} x$$

$$x_1 = \frac{-7.1 \times 10^{-4} + \sqrt{(7.1 \times 10^{-4})^2 - 4 \times 1 \times (-3.55 \times 10^{-4})}}{2 \times 1}$$

$$x_1 = 0.0185$$

$$x_2 = -0.199$$

$$\text{pH} = -\log(0.0185)$$

$$\text{pH} = 1.73$$

pH of Weak Acid/Base Solutions

- Weak acid/base, the change of concentration is very small compared to the initial concentration of acid/base.

Example: Calculate the **pH** of a 0.50 M HF solution at 25 °C.



$$\frac{x^2}{0.5 - x} \approx \frac{x^2}{0.5} = 7.1 \times 10^{-4}$$

$$x = \sqrt{7.1 \times 10^{-4} \times 0.5} = 0.0188$$

$$\text{pH} = -\log(0.0188) = 1.72$$

$$\% \text{ ionization} = \frac{[\text{F}^-]_e}{[\text{HF}]_0} = \frac{0.0188 \text{ M}}{0.50 \text{ M}} \times 100\% = 3.8\% < 5\%$$

- The percent ionization:
- For monoprotic acid:

$$\text{percent ionization} = \frac{\text{ionized acid concentration at equilibrium}}{\text{initial concentration of acid}} \times 100\%$$

$$\frac{[\text{A}^-]_e}{[\text{HA}]_0} \times 100\% =$$

$$\% \text{ ionization} = \frac{[\text{H}^+]_e}{[\text{A}]_0}$$

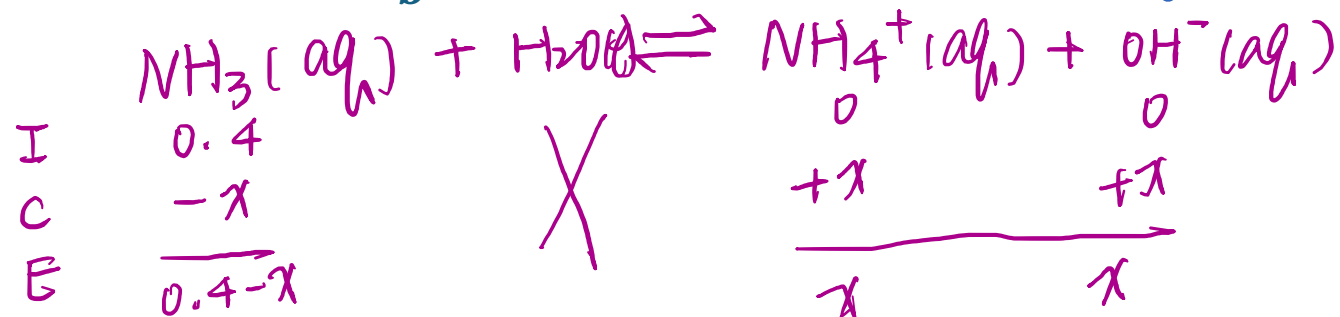
- When percent ionization < 5%, $[\text{HA}]_e \approx [\text{HA}]_0$

Your turn!

What is the pH and percentage ionization of a 0.4 M ammonia solution at 25 °C? The K_b of ammonia is 1.8×10^{-5} .

Your turn!

What is the pH and percentage ionization of a 0.4 M ammonia solution? The K_b of ammonia is 1.8×10^{-5} . ✓



$$\frac{x^2}{0.4-x} \approx \frac{x^2}{0.4} = 1.8 \times 10^{-5} \quad x = 2.7 \times 10^{-3}$$

$$[\text{OH}^-] = 2.7 \times 10^{-3} \text{ M}$$

$$pOH = -\log(2.7 \times 10^{-3}) = 2.57$$

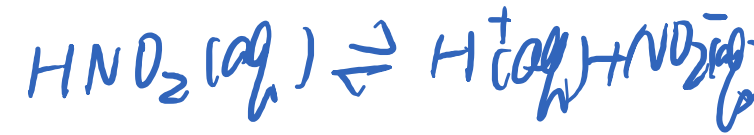
$$pH = 14 - 2.57 = 11.43$$

$$\% \text{ ionization} = \frac{2.7 \times 10^{-3} \text{ M}}{0.4 \text{ M}} \times 100\% = 0.67\% < 5\%$$

Your turn!

Calculate the **pH** and percent ionization of a 0.36 M nitrous acid (HNO_2) solution. The **K_a** of HNO_2 is 4.5×10^{-4} .

$$\frac{x^2}{0.36} = 4.5 \times 10^{-4}$$



$$[\text{H}^+] = x = \sqrt{[\text{HNO}_2]_0 \times K_a}$$

$$\text{pH} = -\log \left(\sqrt{4.5 \times 10^{-4} \times 0.36} \right) = 1.9$$

$$\frac{x}{0.36} \times 100\% = 3.5\%$$

Your turn!

Morphine undergoes ionization in water. It is either a monoprotic acid or monohydric base. In 0.010M Morphine 25 °C, the pH is 10.10. ⁷⁷

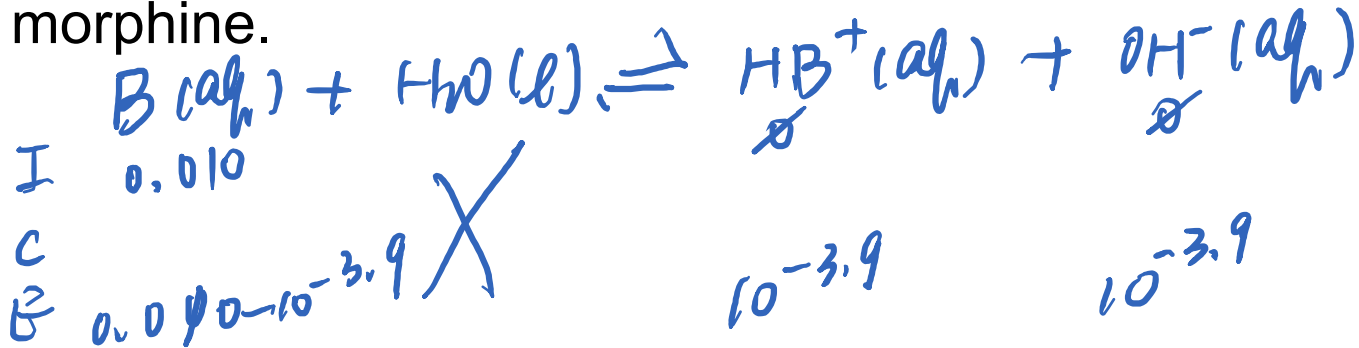
1. This compound is a:

- A. Strong acid
- B. Strong base
- C. Weak acid
- ☒ D. Weak base

Assume it is strong base
 $[OH^-] = 0.010 M = 10^{-2} M$
 $pOH = -\log(10^{-2}) = +2$
 $pH = 14 - 2 = 12 < 10.10$



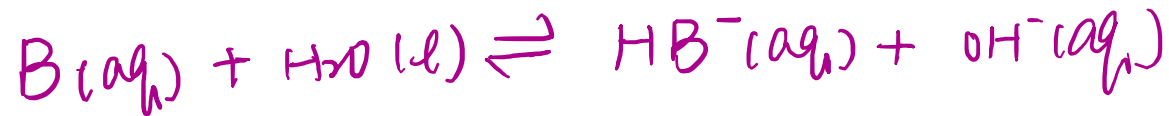
2. Calculate percent ionization and equilibrium constant for the ionization of morphine.



$$[OH^-] = 10^{-(14 - 10.10)} = 10^{-3.9}$$

$$K = \frac{[10^{-3.9}]^2}{0.010 - 10^{-3.9}} = 1.6 \times 10^{-6}$$

Your turn!



Morphine undergoes ionization in water. It is either a monoprotic acid or monohydric base. In 0.010M Morphine, the pH is 10.10.

1. This compound is a:

- A. Strong acid
- B. Strong base
- C. Weak acid
- D. Weak base



2. Calculate percent ionization and equilibrium constant for the ionization of morphine.

$$[OH^-] = 10^{-(14 - 10.10)} = 1.26 \times 10^{-4} M$$

$$\% \text{ ionization} = \frac{1.26 \times 10^{-4} M}{0.010 M} \times 100\% = 1.3\%$$

$$K_b = \frac{(1.26 \times 10^{-4})^2}{0.010 - 1.26 \times 10^{-4}} = 1.6 \times 10^{-6}$$

Announcement 5/6

Assignments that are due soon

- Pre-Class Assignment Week 14-F
- Case study (Turn it in before Fri lecture 5/8)

Chem RePS Session

Wed 2:00 - 3:30 PM at KC 230

2nd Chem RePS or SI worksheets for Unit 4
will be closed on 5/8

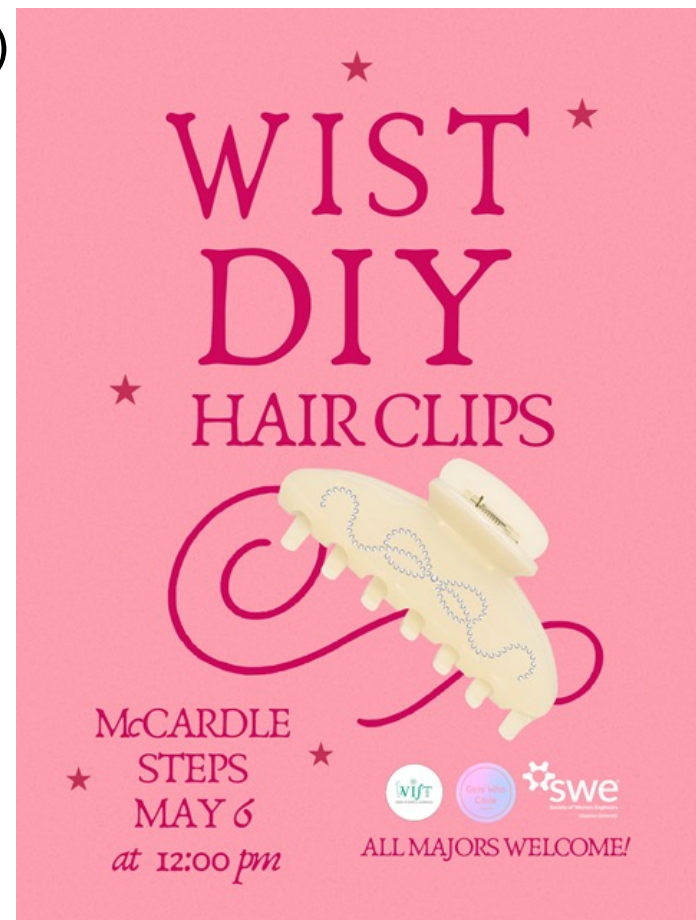
Office Hour

- Dr. Wang's: Thu 10-11 AM on Zoom

SI Session

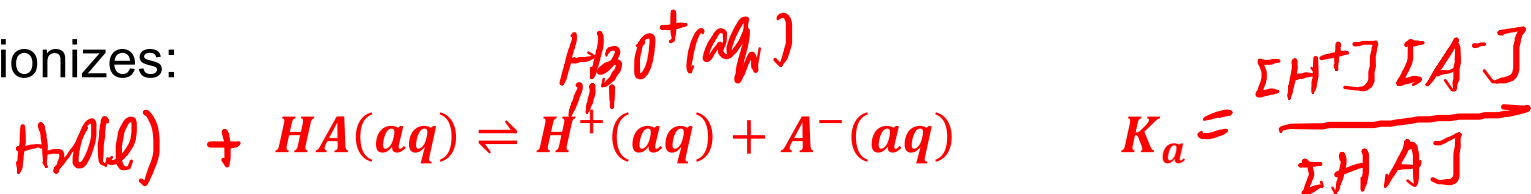
Saige Douglass – Wed 5:30 – 7:30 PM at
Hashinger 412

Paige Kwan - week 14 SI worksheet uploaded

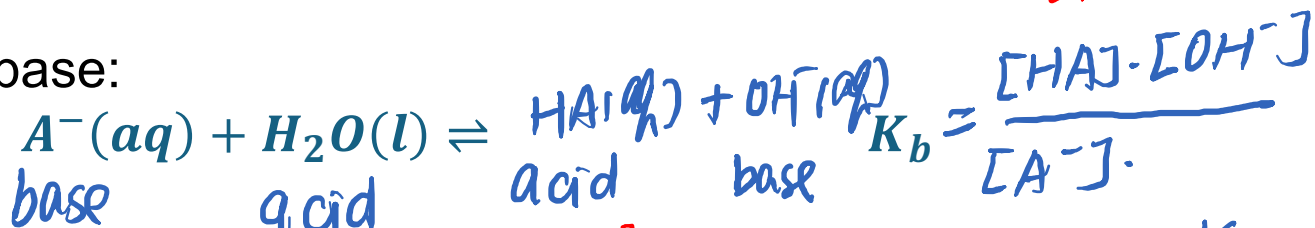


The Relationship Between K_a and K_b

when HA ionizes:



• For its conjugate base:



$$K_a \times K_b = \frac{[H^+][A^-]}{[HA]} \times \frac{[HA] \cdot [OH^-]}{[A^-]} = [H^+] \times [OH^-] = K_w$$

$$\boxed{K_a \times K_b = K_w}$$

• $pK_a = -\log(K_a)$; $pK_b = -\log(K_b)$ $-\log(K_a \times K_b) = -\log(10^{-14})$

$$-14 \quad (-\log K_a) + (-\log K_b) = +14$$

• At 25 °C, $K_w = 10^{-14}$

$$\boxed{pK_a + pK_b = 14}$$

$$\boxed{pH + pOH = 14}$$

The Relationship Between K_a and K_b $K_a \times K_b = 10^{-14}$

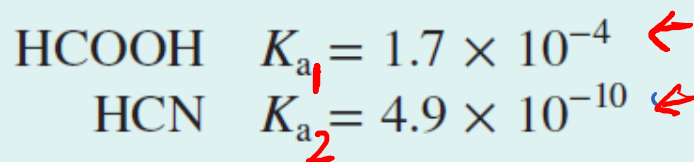
- The **stronger** an acid is, the weaker its conjugate base will be
- The **weaker** an acid is, the stronger its conjugate base will be
- The acid/base reaction favors the direction to form weaker acid/base.

Name of Acid	Formula	K_a	Conjugate Base	K_b^+
Hydrofluoric acid	HF	7.1×10^{-4}	F^-	1.4×10^{-11}
Nitrous acid	HNO_2	4.5×10^{-4}	NO_2^-	2.2×10^{-11}
Acetylsalicylic acid (aspirin)	$C_9H_8O_4$	3.0×10^{-4}	$C_9H_7O_4^-$	3.3×10^{-11}
Formic acid	HCOOH	1.7×10^{-4}	$HCOO^-$	5.9×10^{-11}
Ascorbic acid*	$C_6H_8O_6$	8.0×10^{-5}	$C_6H_7O_6^-$	1.3×10^{-10}
Benzoic acid	C_6H_5COOH	6.5×10^{-5}	$C_6H_5COO^-$	1.5×10^{-10}
Acetic acid	CH_3COOH	1.8×10^{-5}	CH_3COO^-	5.6×10^{-10}
Hydrocyanic acid	HCN	4.9×10^{-10}	CN^-	2.0×10^{-5}
Phenol	C_6H_5OH	1.3×10^{-10}	$C_6H_5O^-$	7.7×10^{-5}

Your turn!



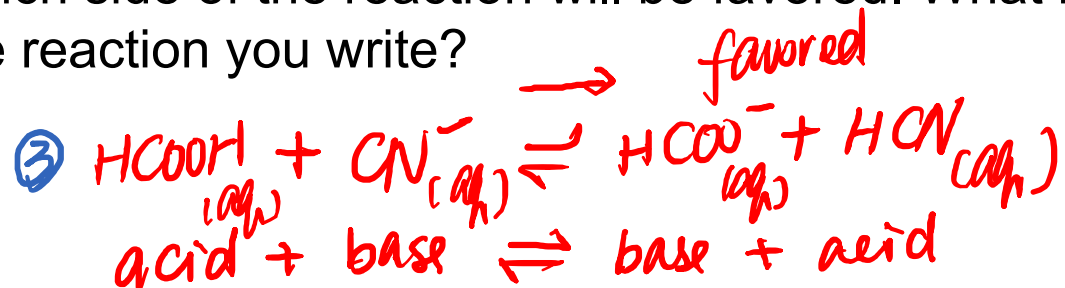
Consider the following two acids and their ionization constants:



Which conjugate base (HCOO^- or CN^-) is stronger?



Write the Bronsted acid/base reaction contains these two acid and predict which side of the reaction will be favored. What is the equilibrium constant of the reaction you write?



$$K = K_{a1} \times (K_{a2})^{-1} = \frac{K_{a1}}{K_{a2}} = \frac{[\text{H}^+][\text{HCOO}^-]}{[\text{HCOOH}]} \times \frac{[\text{HCN}]}{[\text{H}^+][\text{CN}^-]} = 3.5 \times 10^5 \gg 1$$

$$= \frac{[\text{HCOO}^-][\text{HCN}]}{[\text{HCOOH}][\text{CN}^-]} = \frac{1.7 \times 10^{-4}}{4.9 \times 10^{-10}} \uparrow$$